Family Name ....

Given Name ....

Student No. ....

Signature ....

THE UNIVERSITY OF NEW SOUTH WALES

School of Electrical Engineering & Telecommunications

MID-TERM EXAMINATION

Term 3, 2019

ELEC1111
Electrical and Telecommunications Engineering

TIME ALLOWED:	**75 min**
TOTAL MARKS:	**100**
TOTAL NUMBER OF QUESTIONS:	**5**

THIS EXAM CONTRIBUTES 25% TO THE TOTAL COURSE ASSESSMENT

Reading Time: 5 minutes.

This paper contains 5 pages.

Candidates must **ATTEMPT ALL** questions.

Answer each question in a **separate answer booklet**.

Marks for each question are indicated beside the question.

This paper **MAY NOT** be retained by the candidate.

Print your name, student ID and question number on the front page of each answer book.

Authorised examination materials:

Candidates should use their own UNSW-approved electronic calculators.

This is a closed book examination.

Assumptions made in answering the questions should be stated explicitly.

All answers must be written in ink. Except where they are expressly required, pencils **may only be used** for drawing, sketching or graphical work.

QUESTION 1 [15 marks]

a. **(9 marks)** For the circuit shown in Figure 1,

   i. **(7 marks)** Calculate the equivalent resistance R_{eq} as seen from terminals *a*-*b*.

   ii. **(2 marks)** Find current i using the result of part (i).

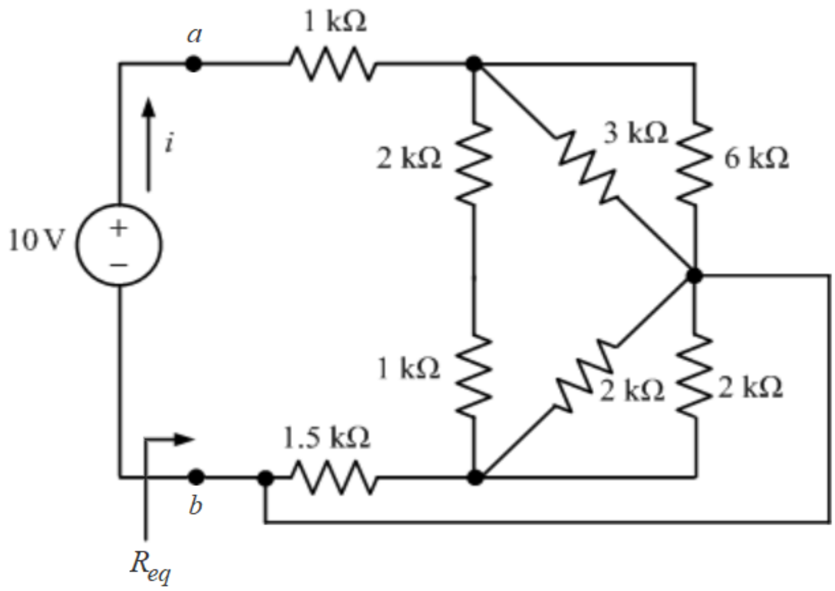

Figure 1

b. **(6 marks)** An electric pencil sharpener rated 240 mW, 6 V is connected to a 9 V battery as shown in Figure 2. Calculate the value of the series-dropping resistor, R_x, needed to power the sharpener.

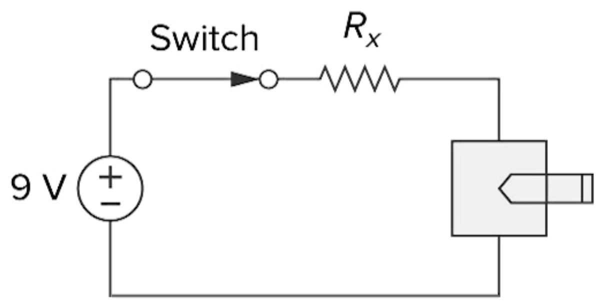

Figure 2

QUESTION 2 [25 marks]

a. **(13 marks)** For the circuit shown in Figure 3,

   i. **(8 marks)** Apply <u>nodal analysis</u> and write down the nodal equations using the labels provided. Note: Simplify the equations, but DO NOT solve them.

   ii. **(5 marks)** Given the values of nodal voltages as $v_1 = 100\ V$, $v_2 = 550\ V$ and $v_3 = 400\ V$, find the power in the $300\ V$ voltage source.

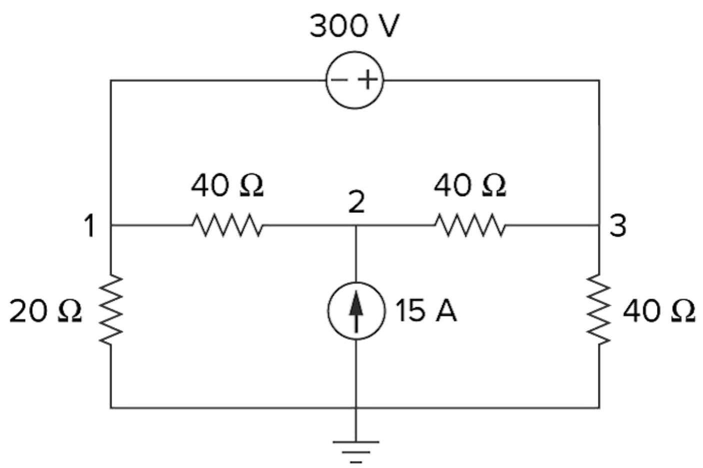

Figure 3

b. **(12 marks)** Bipolar transistors can serve as amplifiers, producing both current gain and voltage gain. The circuitry in the dashed box models a bipolar junction transistor and the whole circuit in Figure 4 models a transistor amplifier.

   For the transistor amplifier shown in Figure 4, apply <u>mesh analysis</u> and write down the mesh equations using the labels provided. Note: Simplify the equations, but DO NOT solve them.

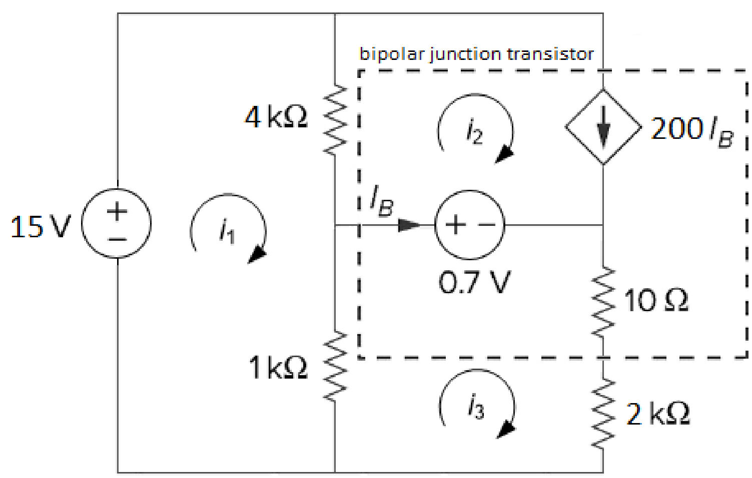

Figure 4

QUESTION 3 [15 marks]

For the circuit shown in Figure 5, use a succession of source transformations to find the voltage v_0.

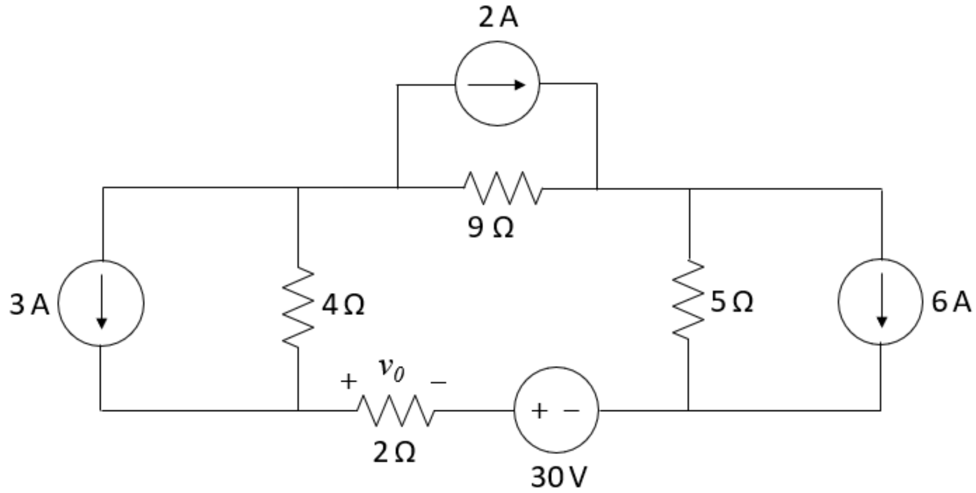

Figure 5

QUESTION 4 [20 marks]

The circuit shown in Figure 6 is being used to power a hot water heater. The heater can be modelled as a single resistor connected to terminals *a*-*b*.

a. (9 marks) Calculate the Thevenin voltage at terminals *a*-*b*.

b. (9 marks) Calculate the heater resistance that will ensure the maximum transfer of power from the circuit to the heater.

c. (2 marks) Find the maximum power that can be delivered to the heater.

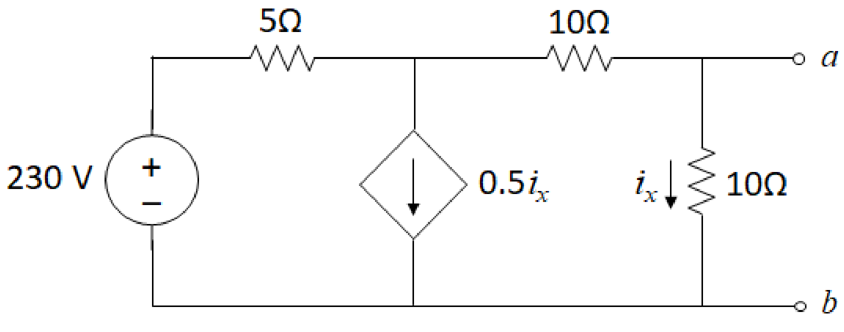

Figure 6

QUESTION 5 [25 marks]

a. **(10 marks)** Find the voltage across the capacitor, v_C , and the current flowing through the inductor, i_L , in the circuit of Figure 7 under DC steady state conditions.

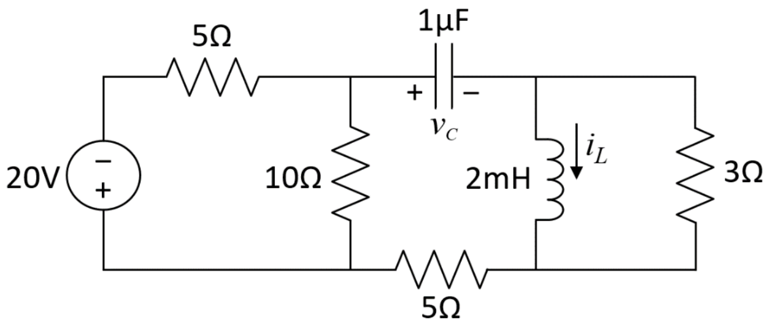

Figure 7

b. **(15 marks)** The circuit in Figure 8 is designed to operate an alarm which activates when the current through it exceeds 0.1 mA.

- **i.** **(10 marks)** Calculate the voltage across the capacitor for $t > 0$ s. Assume that the capacitor was initially discharged.
- **ii.** **(5 marks)** How long does it take for the alarm to activate for the first time after the switch is closed?

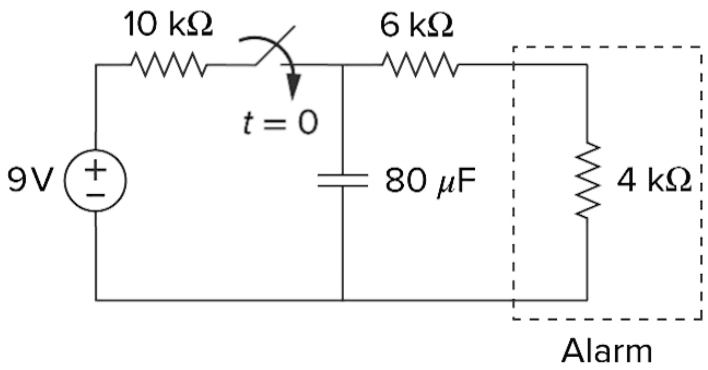

Figure 8